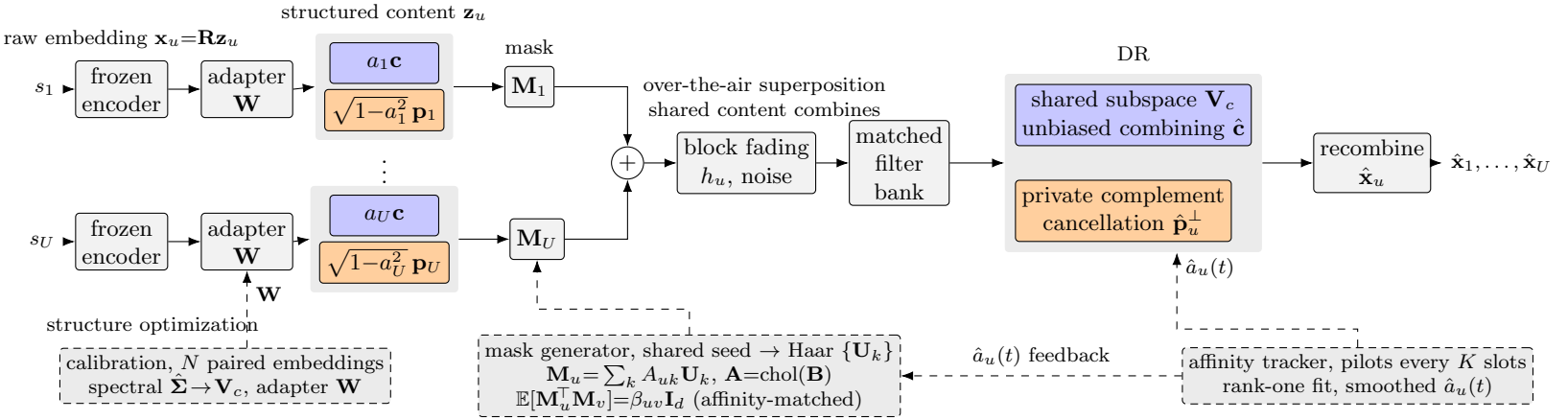

structured content $\mathbf{z}_u$
raw embedding $\mathbf{x}_u$=$\mathbf{R}\mathbf{z}_u$
s_1
frozen encoder
adapter $\mathbf{W}$
$a_1\mathbf{c}$
$\sqrt{1-a_1^2}\,\mathbf{p}_1$
mask
$\mathbf{M}_1$
s_U
frozen encoder
adapter $\mathbf{W}$
$a_U\mathbf{c}$
$\sqrt{1-a_U^2}\,\mathbf{p}_U$
$\mathbf{M}_U$
$\mathbf{W}$
structure optimization
calibration, N paired embeddings
spectral $\hat{\mathbf{\Sigma}}\to\mathbf{V}_c$, adapter $\mathbf{W}$
over-the-air superposition
shared content combines
block fading
h_u, noise
matched filter bank
DR
shared subspace $\mathbf{V}_c$
unbiased combining $\hat{\mathbf{c}}$
private complement cancellation $\hat{\mathbf{p}}_u^{\perp}$
recombine $\hat{\mathbf{x}}_u$
$\hat{\mathbf{x}}_1,\ldots,\hat{\mathbf{x}}_U$
$\hat{a}_u(t)$
mask generator, shared seed → Haar $\{\mathbf{U}_k\}$
$\mathbf{M}_u=\sum_k A_{uk}\mathbf{U}_k$, $\mathbf{A}$=chol($\mathbf{B}$)
$\mathbb{E}[\mathbf{M}_u^{\top}\mathbf{M}_v]=\beta_{uv}\mathbf{I}_d$ (affinity-matched)
$\hat{a}_u(t)$ feedback
affinity tracker, pilots every K slots
rank-one fit, smoothed $\hat{a}_u(t)$